



SIGNAL ANALYSIS REPORT

Automated Signal Analysis and Parameter Extraction
SIH Signal Analyzer • National Technical Research Organisation

Date / Time: 2026-09-24 16:06:49
Input File: synthetic_BPSK_1000kHz_SNR20dB.wav
Format: WAV Audio / IQ (.WAV)

1. FILE / SIGNAL INFORMATION

File Name	synthetic_BPSK_1000kHz_SNR20dB.wav
File Format	WAV Audio / IQ (.WAV)
File Size	69,676 bytes
Signal Duration	17.41 ms (0.01741 s)
Sample Rate (Fs)	1.0000 MHz (1,000,000.0 Hz)
Center Freq Offset	-4.88 Hz
Total Samples	17,408

2. DETECTED SIGNAL PARAMETERS

Modulation	BPSK
Modulation Confidence	100.00%
Occupied Bandwidth	173.828 kHz (173,828.1 Hz)
Carrier Freq Offset (CFO)	-4.88 Hz
Symbol Rate	Not available
Samples Per Symbol	Not available
Frame Sync Status	Unlocked / Direct

3. MODULATION CLASSIFICATION

Modulation	Probability	Distribution
BPSK	100.00%	
QPSK	0.00%	
GMSK	0.00%	

4. TIME-DOMAIN WAVEFORM

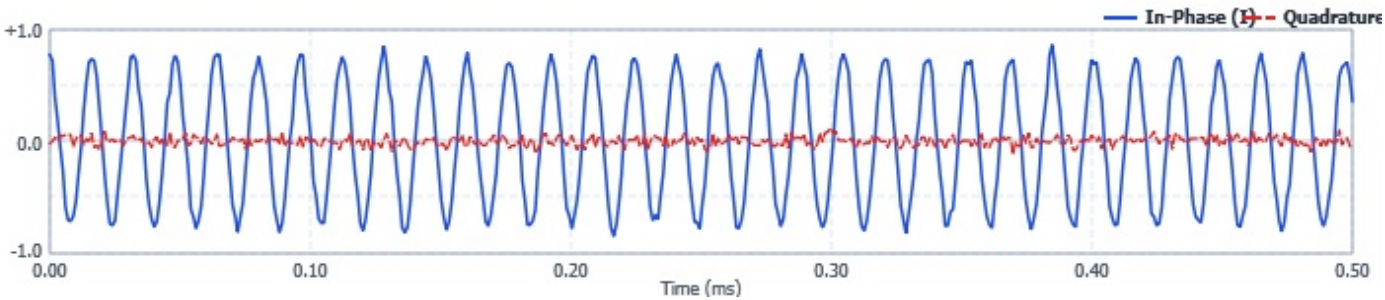


Figure 1: Time-Domain Waveform (In-Phase I & Quadrature Q Components)

5. FREQUENCY SPECTRUM

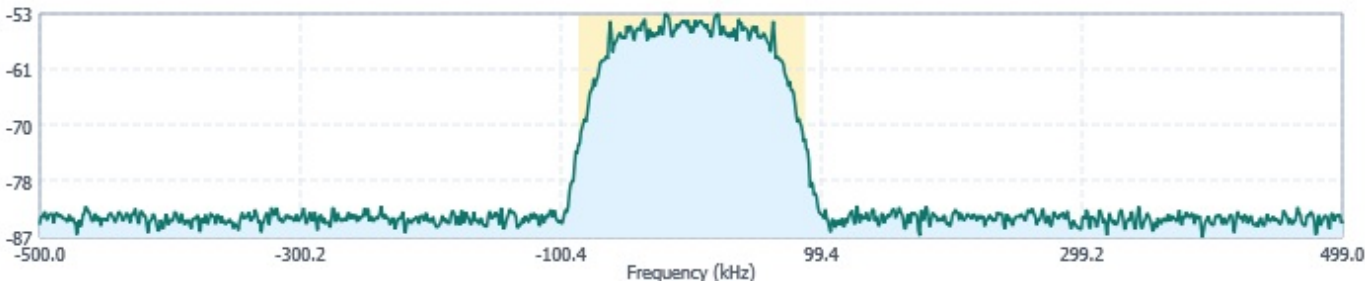


Figure 2: Power Spectral Density (PSD) and Occupied Bandwidth

6. SPECTROGRAM / WATERFALL

7. CONSTELLATION DIAGRAM

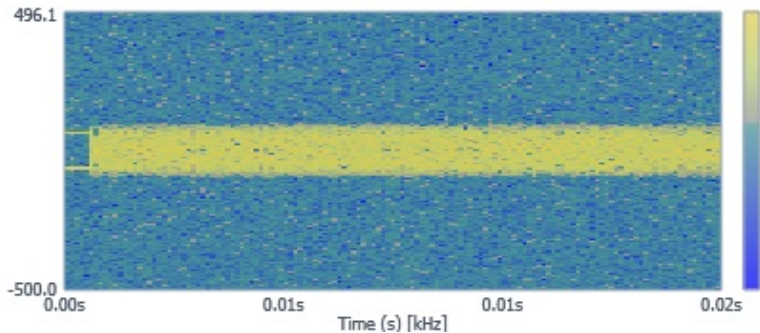


Figure 3: Time-Frequency STFT Spectrogram

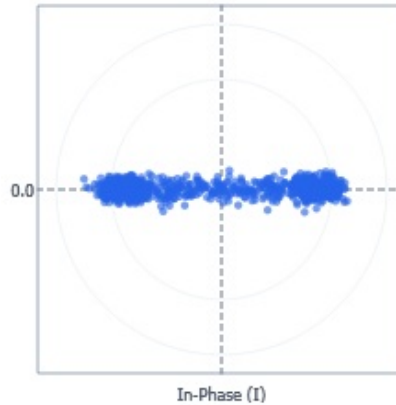


Figure 4: IQ Constellation Diagram

8. SYNCHRONIZATION / FRAME ANALYSIS

Header Offset	0 bits	Payload Offset	0 bits
Correlation Sync Match	Not available	Sync Pattern Locked	Direct framing
Phase Inversion Resolved	No (Direct polarity)	Demodulated Bits Count	128 bits

9. BITSTREAM / DECODING

Total Decoded Bits	128 bits	Decoded Bytes	16 bytes
Bit Density (1s / 0s)	64 (50.0%) / 64 (50.0%)	ASCII Text Preview	BPSK TEST PAYLOAD
Bitstream Sample	01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101		

Decoded Bytes Hex Dump:

000000	42 50 53 4B 20 54 45 53	54 20 50 41 59 4C 4F 41	BPSK TEST PAYLOA
000010	44		D

10. FEC / DE-INTERLEAVER ANALYSIS

Component	Detected Configuration	Status / Confidence
De-interleaver	None	Bypassed
Forward Error Correction (FEC)	None	Uncoded
Channel Decoder	Automated Multi-Scheme Pipeline	16 bytes recovered

11. FINAL ANALYSIS SUMMARY

RF capture file 'synthetic_BPSK_1000kHz_SNR20dB.wav' was processed, containing 17,408 samples recorded at a sample rate of 1.0000 MHz (1,000,000.0 Hz) (effective duration: 17.41 ms (0.01741 s)). The neural modulation classifier identified the transmission format as BPSK with a classification confidence of 100.00%. Spectral analysis determined an occupied bandwidth of 173.828 kHz (173,828.1 Hz) with a carrier frequency offset of -4.88 Hz. Demodulation recovered 128 raw bit transitions. Channel decoding detected direct (no) interleaving, and raw uncoded bit framing, successfully assembling 16 decoded payload bytes.